

Scientific Communities in the Literature Selection

Inductive thematic coding based on article titles and keywords

This report summarizes the scientific communities visible in the literature selection. The input file contains 250 records. Categories were formed from recurring terms and themes in the title and keyword fields only. Because publications often belong to more than one research community, the categories are intentionally non-exclusive; the count column therefore indicates how many records match a community, not a partition of the full dataset.

Interpretation note

The categories are meant as an analytical map of communities rather than final labels for single papers. They can be used as a coding frame for a subsequent manual review.

Overview of generated communities

No.	Community	Matching records	Definition	Example from the dataset
1	STEM, science and mathematics education	45	Research community concerned with learning, teaching, assessment and curriculum in STEM fields, including mathematics, science, engineering, biology, computing and data-related education.	Revisiting barriers to implementation of bioinformatics into life sciences education Keywords: undergraduate education; STEM education; bioinformatics; barriers to integration; education reform
2	Educational technology, online learning and learning analytics	84	Community studying digital infrastructures and tools for education, including online learning, AI, learning analytics, virtual reality, programming environments and technology-enhanced learning.	Automatic feedback and assessment of team-coding assignments in a DevOps context Keywords: automatic feedback; assessment; team-coding assignments; DevOps; programming education; learning analytics
3	Higher education, doctoral education and academic careers	53	Community focused on universities and postsecondary education, including undergraduate learning, doctoral training, academic career pathways, campus experiences and graduate outcomes.	Making strides in doctoral-level career outcomes reporting: a review of classification and visualization methodologies in graduate education Keywords: graduate education; doctoral training; program evaluation; career outcomes; taxonomy; PhD
4	Teacher education, professional development and teacher learning	41	Community examining teachers as learners and professionals, including pre-service and in-service teacher education, professional development, teacher cognition and communities of practice.	Mind the Gap: How Teachers Professional Development Preferences Relate to Scientific Recommendations Keywords: teacher professional development; video vignettes; evidence-based interventions; teacher preferences; implementation
5	Assessment, psychometrics and measurement	62	Community concerned with instruments, tests, scales, rubrics, automated feedback, validity, reliability and the modelling or classification of educational outcomes.	Development and validation of a rapid and precise online sentence reading efficiency assessment Keywords: dyslexia; reading fluency assessment; screening tools; reading fluency and comprehension; progress monitoring; psychometrics
6	Equity, inclusion, diversity and access	42	Community studying unequal access, representation and participation in education, including equity, inclusion, diversity, rurality, first-generation status, gender, race, ethnicity and disability.	Support without Status: Inequities in Student-Advisor Relational Dynamics between First-Generation and Continuing-Generation Doctoral Students Keywords: first-generation students; doctoral education; STEM education; faculty advisor relationships; cultural capital; inequality

No.	Community	Matching records	Definition	Example from the dataset
7	Literacy, language and reading research	52	Community focused on reading, writing, language development, literacy practices, vocabulary, comprehension, textual analysis and language learning.	Measuring Oral Reading Fluency (ORF) Computer-Based and Paper-Based: Examining the Mode Effect in Reading Accuracy and Reading Fluency Keywords: oral reading fluency; computer-based assessment; paper-based assessment; mode effect; reading accuracy; reading fluency
8	Social-emotional learning, wellbeing and motivation	56	Community studying non-cognitive and affective dimensions of education, including wellbeing, mental health, stress, resilience, motivation, engagement, self-efficacy, identity and attitudes.	Risk and Protective Factors of College Students Psychological Well-Being During the COVID-19 Pandemic: Emotional Stability, Mental Health, and Household Resources Keywords: COVID-19 pandemic; college students; psychological well-being; emotions; experience sampling method; personality; intraindividual trajectories
9	COVID-19 and emergency remote teaching	16	Community addressing educational disruption and adaptation during the COVID-19 pandemic, especially remote teaching, lockdown learning, stress, and changed school or university practices.	Lockdown Learning: Changes in Online Foreign-Language Study Activity and Performance of Dutch Secondary School Students During the COVID-19 Pandemic Keywords: distance learning; learning analytics; COVID-19; technology-enhanced learning; secondary education
10	Curriculum, pedagogy and instructional design	44	Community interested in the design of learning environments and instructional approaches, including curriculum reform, pedagogy, classroom practices, interventions and design-based research.	Assessing the correspondence of the basic science curriculum across the different classes with core curriculum specifications in Ebonyi State, Nigeria Keywords: correspondence; basic science curriculum; core curriculum; specifications; secondary schools
11	Collaboration, social networks and communication	29	Community examining learning as a social and networked process, including collaboration, peer relationships, team learning, social media, online communities and science communication.	Promoting collaborative reflection to foster interprofessional collaboration skills: Putting laboratory findings to the test in the field Keywords: collaborative reflection; interprofessional collaboration; higher education; CSCL; technology-enhanced learning
12	Educational policy, program evaluation and school improvement	30	Community oriented toward policy, institutional change, implementation, program evaluation, school improvement, accountability and the effectiveness of educational initiatives.	An Evaluation of the Educational Impact of College Campus Visits: A Randomized Experiment Keywords: college access; experimental research; postsecondary education; program evaluation; rural education; socioemotional skills; survey research
13	Open science, bibliometrics and research synthesis	29	Community concerned with how research is produced, aggregated and made transparent, including systematic reviews, bibliometric mapping, preregistration, reproducibility and open science practices.	Research on the application of ICT in Mathematics education: Bibliometric analysis of scientific bibliography from the Scopus database Keywords: ICT; mathematics education; bibliometric analysis; Scopus
14	Early childhood, primary and secondary schooling	48	Community focused on pre-university settings and learners, including early childhood education, primary and secondary schooling, children, adolescents and school-age populations.	Using the thin slices technique to assess interactional quality in early childhood education and care settings Keywords: thin slices technique; ECEC research; interaction quality; psychometric accuracy; first impressions; applied measurement

Category definitions and coding logic

1. STEM, science and mathematics education (45 matching records)

Definition: Research community concerned with learning, teaching, assessment and curriculum in STEM fields, including mathematics, science, engineering, biology, computing and data-related education.

Example: Revisiting barriers to implementation of bioinformatics into life sciences education

Example keywords: undergraduate education; STEM education; bioinformatics; barriers to integration; education reform

2. Educational technology, online learning and learning analytics (84 matching records)

Definition: Community studying digital infrastructures and tools for education, including online learning, AI, learning analytics, virtual reality, programming environments and technology-enhanced learning.

Example: Automatic feedback and assessment of team-coding assignments in a DevOps context

Example keywords: automatic feedback; assessment; team-coding assignments; DevOps; programming education; learning analytics

3. Higher education, doctoral education and academic careers (53 matching records)

Definition: Community focused on universities and postsecondary education, including undergraduate learning, doctoral training, academic career pathways, campus experiences and graduate outcomes.

Example: Making strides in doctoral-level career outcomes reporting: a review of classification and visualization methodologies in graduate education

Example keywords: graduate education; doctoral training; program evaluation; career outcomes; taxonomy; PhD

4. Teacher education, professional development and teacher learning (41 matching records)

Definition: Community examining teachers as learners and professionals, including pre-service and in-service teacher education, professional development, teacher cognition and communities of practice.

Example: Mind the Gap: How Teachers Professional Development Preferences Relate to Scientific Recommendations

Example keywords: teacher professional development; video vignettes; evidence-based interventions; teacher preferences; implementation

5. Assessment, psychometrics and measurement (62 matching records)

Definition: Community concerned with instruments, tests, scales, rubrics, automated feedback, validity, reliability and the modelling or classification of educational outcomes.

Example: Development and validation of a rapid and precise online sentence reading efficiency assessment

Example keywords: dyslexia; reading fluency assessment; screening tools; reading fluency and comprehension; progress monitoring; psychometrics

6. Equity, inclusion, diversity and access (42 matching records)

Definition: Community studying unequal access, representation and participation in education, including equity, inclusion, diversity, rurality, first-generation status, gender, race, ethnicity and disability.

Example: Support without Status: Inequities in Student-Advisor Relational Dynamics between First-Generation and Continuing-Generation Doctoral Students

Example keywords: first-generation students; doctoral education; STEM education; faculty advisor relationships; cultural capital; inequality

7. Literacy, language and reading research (52 matching records)

Definition: Community focused on reading, writing, language development, literacy practices, vocabulary, comprehension, textual analysis and language learning.

Example: Measuring Oral Reading Fluency (ORF) Computer-Based and Paper-Based: Examining the Mode Effect in Reading Accuracy and Reading Fluency

Example keywords: oral reading fluency; computer-based assessment; paper-based assessment; mode effect; reading accuracy; reading fluency

8. Social-emotional learning, wellbeing and motivation (56 matching records)

Definition: Community studying non-cognitive and affective dimensions of education, including wellbeing, mental health, stress, resilience, motivation, engagement, self-efficacy, identity and attitudes.

Example: Risk and Protective Factors of College Students Psychological Well-Being During the COVID-19 Pandemic: Emotional Stability, Mental Health, and Household Resources

Example keywords: COVID-19 pandemic; college students; psychological well-being; emotions; experience sampling method; personality; intraindividual trajectories

9. COVID-19 and emergency remote teaching (16 matching records)

Definition: Community addressing educational disruption and adaptation during the COVID-19 pandemic, especially remote teaching, lockdown learning, stress, and changed school or university practices.

Example: Lockdown Learning: Changes in Online Foreign-Language Study Activity and Performance of Dutch Secondary School Students During the COVID-19 Pandemic

Example keywords: distance learning; learning analytics; COVID-19; technology-enhanced learning; secondary education

10. Curriculum, pedagogy and instructional design (44 matching records)

Definition: Community interested in the design of learning environments and instructional approaches, including curriculum reform, pedagogy, classroom practices, interventions and design-based research.

Example: Assessing the correspondence of the basic science curriculum across the different classes with core curriculum specifications in Ebonyi State, Nigeria

Example keywords: correspondence; basic science curriculum; core curriculum; specifications; secondary schools

11. Collaboration, social networks and communication (29 matching records)

Definition: Community examining learning as a social and networked process, including collaboration, peer relationships, team learning, social media, online communities and science communication.

Example: Promoting collaborative reflection to foster interprofessional collaboration skills: Putting laboratory findings to the test in the field

Example keywords: collaborative reflection; interprofessional collaboration; higher education; CSCL; technology-enhanced learning

12. Educational policy, program evaluation and school improvement (30 matching records)

Definition: Community oriented toward policy, institutional change, implementation, program evaluation, school improvement, accountability and the effectiveness of educational initiatives.

Example: An Evaluation of the Educational Impact of College Campus Visits: A Randomized Experiment

Example keywords: college access; experimental research; postsecondary education; program evaluation; rural education; socioemotional skills; survey research

13. Open science, bibliometrics and research synthesis (29 matching records)

Definition: Community concerned with how research is produced, aggregated and made transparent, including systematic reviews, bibliometric mapping, preregistration, reproducibility and open science practices.

Example: Research on the application of ICT in Mathematics education: Bibliometric analysis of scientific bibliography from the Scopus database

Example keywords: ICT; mathematics education; bibliometric analysis; Scopus

14. Early childhood, primary and secondary schooling (48 matching records)

Definition: Community focused on pre-university settings and learners, including early childhood education, primary and secondary schooling, children, adolescents and school-age populations.

Example: Using the thin slices technique to assess interactional quality in early childhood education and care settings

Example keywords: thin slices technique; ECEC research; interaction quality; psychometric accuracy; first impressions; applied measurement

Suggested use as a manual coding frame

For manual coding, apply a category when the title or keywords clearly name the community, its object of study, or a characteristic method/topic. Allow multiple codes per publication. If the full paper later contradicts the title/keyword-based assignment, prefer the full-text interpretation.

Most visible communities by keyword/title frequency

Rank	Community	Matching records
1	Educational technology, online learning and learning analytics	84
2	Assessment, psychometrics and measurement	62
3	Social-emotional learning, wellbeing and motivation	56
4	Higher education, doctoral education and academic careers	53
5	Literacy, language and reading research	52
6	Early childhood, primary and secondary schooling	48
7	STEM, science and mathematics education	45
8	Curriculum, pedagogy and instructional design	44
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